Project Charter

A. General Information

Project Title:	**Digital twinning and IoT** **Case study: Elevator**		
Brief Project Description: Write a high-level overview of why you are doing a project. The document explains a project's objectives and its essential qualities. Think of it as the elevator pitch that focuses on what and why without delving into how. The description should focus on goals, objectives, and the overall approach, but you don't need to include lists of tasks, an extensive background, or research analysis. In general, the project description is broad; you will include more detail in the project plan. The project team leader often writes the project description.			
This project is aimed at developing an Industry 4.0 system based on Internet of things. The case study is an elevator system using an IoT platform for data collection and visualization on a web platform. The system is designed to show a real-time working model of the elevator on a webpage and facilitate the control of the physical system remotely. The project can be used to provide students and researchers interested in IoT with the know-how of how it can be used in industries and manufacturing plants. The project can be implemented in industries to provide a remote way of controlling elevators transporting goods to different floors of the industry.			
Prepared By:	Benson Githutha Gathitu E022-01-1071/2018 Frankline Mutuma Mburugu E022-01-1063/2018		
Date:	07/04/2022	***Version:***	1

B. Project Main Objective and Specific objectives:

Write the main objectives and explain the specific objectives of the project. For example: What value does this project add to the organization? How does this project align with the strategic priorities of the organization? What results are expected? What are the deliverables? What benefits will be realized? What problems will be resolved?

Main Objective: - Design of an elevator model digital twin and a web platform that will not only host the digital twin but control the physical elevator and its digital twin simultaneously.

Specific objectives:

1. Design of a digital elevator model using CAD Software
2. Design of a web platform for the control and visualization of the Digital Twin
3. Establish communication between the web platform, microcontroller and database with the CAD design model.
4. Control of the physical elevator using a microcontroller.
5. Control of the digital elevator model and physical elevator simultaneously using a web platform.

C. Assumptions

List and describe the assumptions made in the decision to charter this project. Please note that all assumptions must be validated to ensure that the project stays on schedule and on budget.

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Our project is based on the following assumptions:

- The mechanical parts of the lift structure are in good condition and it will function with ease.
- All the resources required to be used in the project are available, in good condition and ready for use.

D. Project Scope

Describe the scope of the project. The project scope establishes the boundaries of the project. It identifies the limits of the project and defines the deliverables.

This project involves collection of data from a physical elevator system, sending the data through Remote Procedure Call to an API and visualizing the data using a web platform. It also includes the creation of a cloud based digital twin of the elevator that simulates the actual elevator system in real time.

List any requirements that are specifically excluded from the scope.

E. Project Milestones

List the major milestones and deliverables of the project.

Milestones	Deliverables	Date
Design of a digital elevator model using CAD Software	A replica of the physical model that can be visualized and animated will be realized.	May/2022
Design of a web platform for the control and visualization of the Digital Twin	A platform for visualization and control of the digital elevator model will be realized.	May/2022
Establish communication between the web platform, microcontroller and database storing the CAD design model	Communication between microchip, website and the database that will store the digital elevator model will be achieved.	May/2022
Control of the physical elevator using a microcontroller	Floor-wise movement of the physical elevator using a microcontroller will be achieved.	September/2022

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Control of the digital elevator model and physical elevator simultaneously using a web platform	Integration of all systems to achieve a cloud based digital twin that works in synchrony with the physical elevator will be realized.	September/ 2022

F. Impact Statement

List the impact this project may have on existing systems or units.

Potential Impact	Systems / Units Impacted
1. Virtualization of factory operations: Engineers and designers can develop a virtual digital system to be used in a factory, then customize, alter, and make changes without affecting the physical factory operations.	Design centres or Research and Development centres in industries.
2. Modularity of industrial functions or operations.	Industries' plant layouts
3. Remote Capability: The project will be a proof of concept that manufacturing processes involving data collection, monitoring and maintenance can be achieved remotely and in real time.	Flexibility of manually controlled systems as they will will be customised to be operated remotely. Learning Institutions; Students and researchers interacting with and learning the system.

G. Roles and Responsibilities

Describe the roles and responsibilities of project team members followed by the names and contact information for those filling the roles. The table below gives some generic descriptions. Modify, overwrite, and add to these examples to accurately describe the roles and responsibilities for this project.

Project Supervisor: Provides overall direction on the project. Responsibilities include: approve the project charter and plan; secure resources for the project; confirm the project's goals and objectives; keep abreast of major project activities; make decisions on escalated issues; and assist in the resolution of roadblocks.	
Name	**Email / Phone**
Prof. Jean Bosco Byiringiro	+254716150627
Project Team leader: Leads in the planning and development of the project; manages the project to scope. Responsibilities include: develop the project plan; identify project deliverables; identify risks and develop risk management plan; direct the project resources (team members); scope control and change management; oversee quality assurance of the project management process; maintain all documentation including the project plan; report and forecast project status; resolve conflicts within the project or between cross-functional teams; ensure that the project's product meets the business objectives; and communicate project status to stakeholders.	
Name	**Email / Phone**
Benson Githutha Gathitu	bengathitu@gmail.com
Team Member: Works toward the deliverables of the project. Responsibilities include: understand the work to be completed; complete research, data gathering, analysis, and documentation as outlined in the project plan; inform the project manager of issues, scope changes, and risk and quality concerns; proactively communicate status; and manage expectations.	
Name	**Email / Phone**
Frankline Mutuma Mburugu	mburugufrankline1@gmail.com
Ephantus Muturi Mukuri	mukurimuturi@gmail.com

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Mary Wanjeri	maryshiks53@gmail.com
Victor Kiprono Kirui	victorkipronokirui2@gmail.com
Project Coordinator: The person or department requesting the deliverable. Responsibilities include: partner with the project supervisor or team leader to create the Project Charter; partner with the project team leader to manage the project including the timeline, work plan, testing, resources, training, and documentation of procedures; work with the project team to identify the technical approach to be used and the deliverables to be furnished at the completion of the project; provide a clear definition of the business objective; sign-off on project deliverables; take ownership of the developed process and software.	
Name	**Email / Phone**
Prof. Jean Bosco Byiringiro	+254716150627
Subject Matter Expert: Provides expertise on a specific subject. Responsibilities include: maintain up-to-date experience and knowledge on the subject matter; and provide advice on what is critical to the performance of a project task and what is nice-to-know (Maximum two experts)	
Name	**Email / Phone**
Andrew Kibor	andrew.chesang@dkut.ac.ke

H. Resources

Identify the initial funding, personnel, and other resources committed to this project by the project coordinator.

Resource	Constraints
Project Budget	Kshs. 4550
Materials	2 Limit Switches 1 Terminal Block 3 ON-OFF Momentary Push buttons 2 Emergency stop button 1 Custom PCB 1 Single Core Copper wire 1 Solder Paste Flux 1 LCD 1 H-Bridge motor driver
Other Resources	PCs, design software, Mechanical elevator model, NodeMCU microcontroller
Personnel	Benson, Frankline, Ephantus,Mary,Victor

I. Project Risks

Identify the high-level project risks and the strategies to mitigate them.

Risk	Mitigation Strategy
Electronic components used might fail.	Follow manufacturer's standards and specifications or replacement of components if damaged.
Latency in cloud based digital twin	Come up with optimized APIs and using fast internet
Failure of completion of work allocated to a group member due to unforeseen circumstances for example health issues	Acceleration of modules being worked by other group members in order to work on the module left by the group member

J. Success Measurements

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Identify metric and target you are trying to achieve as a result of this project. The overall cost of each project should be **KShs. 50,000** and implemented in a period of eight (**8) months**). For example, overall cost savings of **KShs 30,000** or reduce processing time by **25 percent**.

A cloud based digital twin of the elevator visualized on a web platform working in real-time with the physical one.

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K. Signatures

The signatures of the people below document approval of the formal Project Charter. The project manager is empowered by this charter to proceed with the project as outlined in the charter.

Team Members		
Name	**Signature**	**Date**
Frankline Mutuma Mburugu		07/04/2022
Ephantus Muturi Mukuri		07/04/2022
Mary Wanjeri		07/04/2022
Victor Kiprono Kirui		07/04/2022
Team leader		
Name	Signature	Date
Benson Githutha Gathitu		07/04/2022
Project supervisor		
Name	Signature	Date
Prof. Jean Bosco Byiringiro		
Project Coordinator		
Name	Signature	Date
Prof. Jean Bosco Byiringiro		